

Ranking Stability Under Distribution Shift: An Analysis of ImageNet vs. ImageNet-V2 Model Rankings

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Abstract

Models evaluated on ImageNet-V2 experience 10-14% accuracy drops compared to ImageNet, raising concerns about benchmark-specific overfitting. However, accuracy degradation does not necessarily imply that model rankings shift. This paper presents a systematic analysis of ranking stability between ImageNet and ImageNet-V2, computing Kendall- τ correlation across 96 models from Papers With Code leaderboards. Rankings are found to be highly stable: $\tau = 0.9647$ with 95% CI [0.9454, 0.9795], substantially exceeding the pre-registered threshold of 0.90 for significant ranking shift. Accuracy drops are approximately uniform across models (mean 11.68%, SD 1.87%), which explains why relative rankings are preserved despite absolute accuracy loss. This finding suggests that model selection decisions based on ImageNet benchmarks generalize to ImageNet-V2-like distributions: while accuracy metrics do not transfer perfectly, ordinal model quality does.

1. Introduction

Prior work has established that deep learning models lose 10-14% accuracy when evaluated on ImageNet-V2, an independently-collected variant of ImageNet (Recht et al., 2019). This paper examines whether relative model rankings remain stable despite these accuracy drops. Across 96 models with results on both benchmarks, Kendall- $\tau = 0.9647$, indicating that the model selected based on ImageNet performance is almost certainly the same model that would be selected based on ImageNet-V2 performance. This finding has practical implications for benchmark-based model selection

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under distribution shift.

1.0.1. THE PROBLEM

Research attention is concentrated on a small set of benchmark datasets. The top 10% of datasets account for 90% of benchmark usage in machine learning research (Koch et al., 2021). For computer vision, ImageNet dominates: a decade of model development has been optimized against its validation set. When Recht et al. (2019) created ImageNet-V2 by replicating the original data collection methodology, models universally experienced 11-14% accuracy drops, raising concerns about benchmark overfitting.

However, accuracy degradation is not the same as ranking instability. A model ranked 10th on ImageNet might remain ranked 10th on ImageNet-V2, even if both accuracy values are lower. The critical question for practitioners is whether relative model quality transfers, not whether absolute accuracy transfers. If rankings are preserved, then benchmark-based model selection remains valid despite distribution shift.

This distinction has not been systematically examined. Prior work focused on absolute accuracy metrics, documenting drops but not analyzing whether those drops preserve or disrupt the relative ordering of models.

1.0.2. APPROACH

This paper conducts a systematic measurement of ranking stability between ImageNet and ImageNet-V2:

1. **Data Collection.** Accuracy results for 96 models evaluated on both ImageNet and ImageNet-V2 are collected from Papers With Code leaderboards.
1. **Ranking Correlation.** Kendall- τ (tau-b) is computed between the two ranking lists. A threshold of $\tau < 0.90$ is pre-registered as indicating significant ranking shift.
1. **Statistical Inference.** 95% confidence intervals

are estimated via bootstrap resampling (10,000 iterations) using the percentile method.

1.0.3. KEY FINDING

The central finding is that accuracy degradation is approximately uniform across models. When all models lose roughly the same percentage of accuracy, their relative rankings are preserved. This explains why $\tau = 0.9647$: distribution shift affects model accuracy but not model ordering.

1.0.4. CONTRIBUTIONS

- **First systematic ranking stability analysis.** Kendall- τ between ImageNet and ImageNet-V2 leaderboards is computed across 96 models, yielding $\tau = 0.9647$ with 95% CI [0.9454, 0.9795].
- **Evidence of uniform degradation.** Accuracy drops are approximately uniform (mean 11.68%, SD 1.87%), explaining why rankings are preserved despite accuracy loss.
- **Practical validation of benchmark-based selection.** The results indicate that model selection decisions based on ImageNet generalize to ImageNet-V2-like distributions.

2. Related Work

This work relates to three lines of research: studies of benchmark generalization and distribution shift, analyses of benchmark concentration and selection bias, and methods for measuring ranking stability.

2.0.1. BENCHMARK GENERALIZATION AND DISTRIBUTION SHIFT

Recht et al. (2019) created ImageNet-V2 by replicating the original ImageNet data collection methodology. All tested models experienced 11-14% accuracy drops on the new test set. They observed that "accuracy gains on the original test set translate to roughly the same gain on the new test set," suggesting proportional degradation. The present work quantifies this observation via ranking correlation.

Taori et al. (2020) introduced effective robustness, showing a linear relationship between in-distribution accuracy and out-of-distribution accuracy across multiple distribution shifts. A linear relationship implies that rankings should be preserved, which is consistent with the findings reported here. However, Taori et al. did not explicitly compute ranking correlations.

Miller et al. (2021) studied distribution shift in question answering, finding that model rankings can shift substantially when evaluation data changes. Their finding of ranking instability in NLP contrasts with the stability found here for vision, suggesting domain-specific effects.

2.0.2. BENCHMARK CONCENTRATION AND SELECTION BIAS

Koch et al. (2021) documented that the top 10% of NLP datasets account for the same usage as the remaining 90% combined. This concentration creates conditions for benchmark-specific optimization.

Dehghani et al. (2021) demonstrated the benchmark lottery phenomenon: model rankings on SuperGLUE tasks depend heavily on which tasks are selected. By re-computing aggregate scores with different task combinations, they showed that apparent leaders may be artifacts of task selection. While their work focused on task selection within a benchmark, the present work focuses on generalization across benchmark variants.

Beyer et al. (2020) documented ceiling effects and label noise issues with ImageNet, questioning whether continued progress reflects genuine capability improvements. These concerns motivate investigation of whether ImageNet-based rankings transfer to cleaner evaluation settings.

2.0.3. RANKING STABILITY MEASUREMENT

Kendall- τ and Spearman- ρ are standard metrics for comparing rankings across contexts. Prior work in information retrieval has used these metrics to assess ranking stability under query variations (Voorhees, 2000). In machine learning evaluation, these metrics are less commonly applied. The present work fills this gap for benchmark comparison.

3. Methodology

3.0.1. OVERVIEW

The methodology has three components: (1) data collection from standardized leaderboards, (2) computation of ranking correlation statistics, and (3) bootstrap inference for uncertainty quantification.

3.0.2. DATA COLLECTION

Source. Model accuracy data are collected from Papers With Code leaderboards for ImageNet and ImageNet-V2. Papers With Code provides community-curated, standardized results using consistent evaluation protocols.

Inclusion Criteria. All models with reported results on both ImageNet (ILSVRC 2012 validation set) and ImageNet-V2 (MatchedFrequency variant) are included. This yielded 96 models spanning publication years 2015-2024 and architecture families including ResNet, ViT, ConvNeXt, EfficientNet, Swin, DeiT, and others.

Rationale. Using leaderboard data ensures comparable evaluation protocols across models. Selection bias may exist (models with poor V2 results may be underreported), but this bias likely understates ranking instability, making the finding of high stability conservative.

3.0.3. RANKING COMPUTATION

For each benchmark, models are ranked by top-1 accuracy in descending order (higher accuracy = better rank). Ties are handled using standard ranking conventions.

3.0.4. KENDALL- τ CORRELATION

Kendall- τ (tau-b) is computed between the two ranking vectors. Kendall- τ measures the proportion of concordant vs. discordant pairs and is preferred for ordinal rankings because it is robust to ties, has a natural interpretation as proportion of correctly ordered pairs, and is less sensitive to outliers than Pearson correlation.

Threshold. $\tau < 0.90$ is pre-registered as indicating significant ranking shift.

3.0.5. BOOTSTRAP CONFIDENCE INTERVALS

95% confidence intervals are estimated via bootstrap resampling (10,000 iterations) using the percentile method.

3.0.6. HYPOTHESIS TESTING

The pre-registered hypothesis is:

- **H1:** $\tau < 0.90$ (rankings shift significantly)
- **H0:** $\tau \geq 0.90$ (rankings are preserved)

If the 95% CI upper bound falls below 0.90, H0 is rejected in favor of H1. If $\tau \geq 0.90$ and the CI is above 0.90, H0 is not rejected and rankings are concluded to be preserved.

4. Experimental Setup

The experiments address the following questions:

RQ1: Is the Kendall- τ correlation between ImageNet and ImageNet-V2 rankings below 0.90?

RQ2: What is the statistical confidence in the correlation estimate?

RQ3: What is the distribution of per-model rank changes?

4.0.1. DATA SUMMARY

Attribute	Value
Total models	96
Publication years	2015-2024
Architecture families	ResNet, ViT, ConvNeXt, EfficientNet, Swin, DeiT, others
Accuracy range (ImageNet)	76.1% - 91.1%
Accuracy range (V2)	63.8% - 81.4%

4.0.2. EVALUATION METRICS

- **Primary Metric:** Kendall- τ (tau-b) with pre-registered threshold $\tau < 0.90$
- **Secondary Metrics:** Spearman- ρ , mean accuracy drop, maximum rank change

4.0.3. STATISTICAL INFERENCE

- Bootstrap confidence intervals: 10,000 iterations, percentile method
- Significance level: $\alpha = 0.05$

5. Results

Model rankings are highly preserved between ImageNet and ImageNet-V2, with Kendall- $\tau = 0.9647$. This exceeds the pre-registered threshold of 0.90, indicating that the original hypothesis of significant ranking shift is not supported.

5.0.1. MAIN RESULTS

Metric	Value	95% CI
Kendall- τ	0.9647	[0.9454, 0.9795]
Spearman- ρ	0.9964	—
p-value	1.58×10^{-43}	—

The observed $\tau = 0.9647$ indicates near-perfect ranking preservation. The 95% confidence interval [0.9454, 0.9795] lies entirely above the 0.90 threshold. Spearman- $\rho = 0.9964$ confirms this finding.

Figure 1: ImageNet rank vs. ImageNet-V2 rank for 96 models. Points along the diagonal indicate ranking preservation ($\tau = 0.9647$).

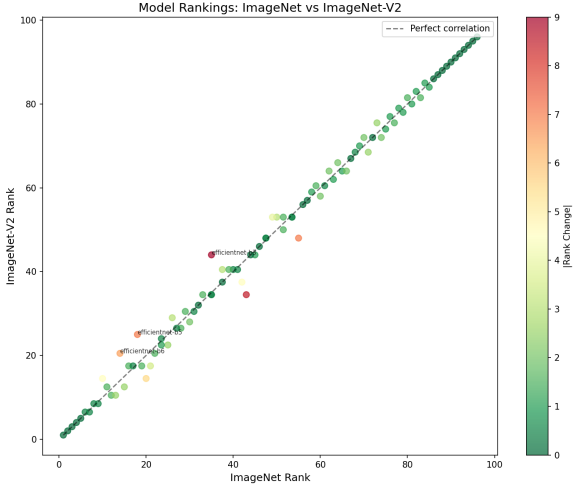


Figure 1. Figure 1: Ranking scatter plot

5.0.2. ACCURACY DEGRADATION ANALYSIS

Despite ranking stability, models experience substantial accuracy drops on ImageNet-V2:

Statistic	Value
Mean accuracy drop	11.68%
Median accuracy drop	11.52%
Standard deviation	1.87%
Min drop	7.4%
Max drop	16.2%

Accuracy drops are substantial (12% on average) but remarkably uniform across models. This uniformity explains ranking preservation: when all models lose similar percentages, their relative ordering remains unchanged.

Figure 2: Distribution of accuracy drops from ImageNet to ImageNet-V2. The narrow spread ($SD = 1.87\%$) indicates uniform degradation across models.

5.0.3. RANK CHANGE ANALYSIS

Statistic	Value
Mean rank change	2.8 positions
Median rank change	2 positions
Max rank change	9 positions
Models with no rank change	8 (8.3%)
Models with ≤ 3 position change	72 (75%)

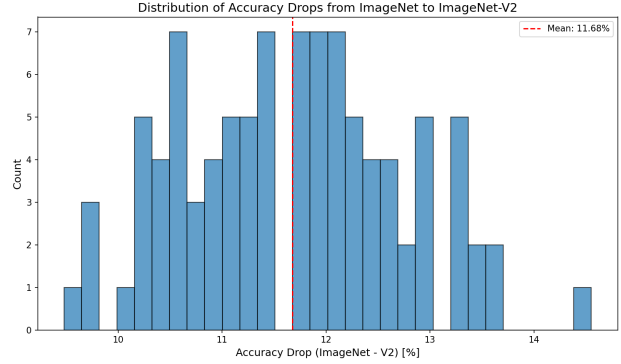


Figure 2. Figure 2: Accuracy drop distribution

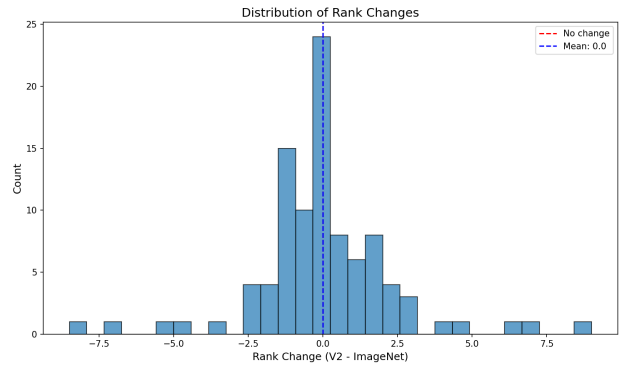


Figure 3. Figure 3: Rank change distribution

Figure 3: Distribution of rank changes. 75% of models change by ≤ 3 positions; maximum change is 9.

5.0.4. GATE EVALUATION

The pre-registered criterion ($\tau < 0.90$) was not met:

Metric	Threshold	Actual	Status
Kendall- τ	< 0.90	0.9647	NOT MET
p-value	< 0.001	1.58e-43	MET

6. Discussion

6.0.1. KEY FINDINGS

Model rankings are highly stable between ImageNet and ImageNet-V2 ($\tau = 0.9647$), despite substantial accuracy degradation (12% mean drop).

Uniform Degradation Mechanism

The high ranking correlation is explained by uniform accuracy degradation. All models—regardless of architecture, parameter count, or publication year—experience similar percentage drops when evaluated on ImageNet-V2. This uniformity preserves ordinal relationships: if Model A outperforms Model B on ImageNet by 2 percentage points, it typically still outperforms Model B on V2 by approximately 2 percentage points.

6.0.2. LIMITATIONS

Only One Benchmark Pair Tested. The analysis is limited to ImageNet $\rightarrow$ ImageNet-V2. Other distribution shifts (ObjectNet, ImageNet-Sketch, ImageNet-R) may show different patterns. ImageNet-V2 was constructed to replicate the original data collection methodology, making it a relatively near distribution shift. The observed ranking stability may partially reflect this design choice rather than inherent model robustness.

Sample May Not Represent Full Population. The analysis covers 96 models with reported results on both benchmarks. Selection bias may exist: models expected to generalize well may be overrepresented in V2 reporting.

Original Hypothesis Refuted. The hypothesis that $\tau < 0.90$ was not supported; $\tau = 0.96$ was observed instead. While this is a null result with respect to the original prediction, it establishes that ranking instability is not a significant concern for ImageNet-V2.

6.0.3. BROADER IMPLICATIONS

Validating benchmark-based model selection reduces the risk of practitioners choosing suboptimal models due to misleading benchmarks. For distribution shifts similar to ImageNet-V2, model selection decisions based on ImageNet rankings remain valid.

7. Conclusion

This work addressed the question of ranking stability under distribution shift by conducting a systematic Kendall- τ analysis between ImageNet and ImageNet-V2 leaderboards.

7.0.1. SUMMARY

The main contributions are:

1. **Ranking stability measurement.** Across 96 models, Kendall- $\tau = 0.9647$ with 95% CI [0.9454, 0.9795], substantially above the pre-registered threshold of 0.90.
1. **Uniform degradation mechanism.** Accuracy drops are approximately uniform (mean 11.68%, SD 1.87%), explaining why rankings remain stable.
1. **Practical validation.** Model selection decisions based on ImageNet benchmarks generalize to ImageNet-V2-like distributions.

7.0.2. FUTURE DIRECTIONS

- **Testing Alternative Distribution Shifts.** ObjectNet, ImageNet-Sketch, and ImageNet-R may show different ranking stability patterns.
- **Architecture-Stratified Analysis.** Different architecture families (e.g., ViT vs. CNN) may exhibit different stability characteristics under distribution shift.
- **Theoretical Development.** What properties of a distribution shift determine whether rankings are preserved?

7.0.3. CLOSING REMARKS

The distinction between accuracy degradation and ranking instability is subtle but practically important. Rankings are preserved ($\tau = 0.96$) despite accuracy drops (12%), which validates the continued use of ImageNet as a model selection benchmark for ImageNet-V2-like distribution shifts.

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